07_final_figures

April 24, 2026

1 Step 7 — Publication-Quality Figures (notebooks/07_final_figures.ipynb)

This notebook assembles the IEEE paper's hero figures and the master results table. It consumes, without recomputing, the outputs of Steps 2–6 stored under `experiments/results/` and the per-step figures already exported to `report/figures/`.

Inputs (read-only):

- `experiments/results/{pd,hd,als}_results.json` — Step 2 within-condition.
- `experiments/results/cross_condition_results.json` — Step 3 transfer.
- `experiments/results/shap_results.json` — Step 4 SHAP j.
- `experiments/results/noise_robustness.json` — Step 5 (optional panels).
- `data/processed/gait_features.csv` + `data/processed/control_partition.json` — used only to reproduce the Step 6 PCA fit for unified styling in Fig 5.

Outputs:

- Six paper figures under `report/figures/paper/{pdf,png}/` at IEEE single-column (3.5 in) or double-column (7.16 in) width: `fig1_within_overview`, `fig2_within_cms`, `fig3_cross_degradation`, `fig4_shap_delta_j`, `fig5_pca_kmeans`, `fig6_noise_robustness`.
- `report/tables/master_results.csv` and `report/tables/master_results.tex` (booktabs-style IEEE `tabular`).

No upstream artefact is modified. Missing optional inputs (Step 5 cross-noise sweep) are skipped with a clear warning rather than failing the notebook.

[1]:
```
import sys
import json
import warnings
from pathlib import Path

sys.path.insert(0, str(Path('..').resolve()))

import numpy as np
import pandas as pd
import polars as pl
import matplotlib
import matplotlib.pyplot as plt
```

```python
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.metrics import confusion_matrix

from src.features import ALL_FEATURE_COLS

IEEE_SINGLE = 3.5
IEEE_DOUBLE = 7.16

matplotlib.rcParams.update({
    'font.family': 'serif',
    'font.size': 9,
    'axes.labelsize': 9,
    'axes.titlesize': 9,
    'xtick.labelsize': 8,
    'ytick.labelsize': 8,
    'legend.fontsize': 7,
    'figure.dpi': 180,
    'savefig.bbox': 'tight',
    'pdf.fonttype': 42,
    'ps.fonttype': 42,
})

PROCESSED = Path('../data/processed')
RESULTS = Path('../experiments/results')
PAPER_PDF = Path('../report/figures/paper/pdf')
PAPER_PNG = Path('../report/figures/paper/png')
TABLES = Path('../report/tables')
for d in (PAPER_PDF, PAPER_PNG, TABLES):
    d.mkdir(parents=True, exist_ok=True)

CLASSIFIERS = ['rf', 'knn', 'svm', 'dt', 'qda', 'xgb', 'lgbm']
CLF_LABEL = {
    'rf': 'RF', 'knn': 'KNN', 'svm': 'SVM', 'dt': 'DT',
    'qda': 'QDA', 'xgb': 'XGB', 'lgbm': 'LGBM',
}
CONDITIONS = ['pd', 'hd', 'als']
COND_LABEL = {'pd': 'PD', 'hd': 'HD', 'als': 'ALS'}
COND_COLORS = {'pd': '#1f77b4', 'hd': '#2ca02c', 'als': '#ff7f0e'}
DIRECTIONS = ['pd_to_hd', 'hd_to_pd', 'pd_to_als', 'als_to_pd', 'hd_to_als',␣
 ↪'als_to_hd']
DIR_LABEL = {d: d.replace('_to_', '→').upper() for d in DIRECTIONS}

def save_paper(fig, stem):
    pdf = PAPER_PDF / f'{stem}.pdf'
```

```
    png = PAPER_PNG / f'{stem}.png'
    fig.savefig(pdf)
    fig.savefig(png, dpi=300)
    print(f'  wrote {pdf.relative_to(Path("..")) } and {png.relative_to(Path("..
↪"))}')

print('Setup complete.')
```

```
Setup complete.
```

[2]:
```
def _load(path: Path, required: bool = True):
    if path.exists():
        with open(path) as f:
            return json.load(f)
    if required:
        raise FileNotFoundError(f'Required input missing: {path}')
    warnings.warn(f'Optional input missing: {path}')
    return None

within_results = {
    c: _load(RESULTS / f'{c}_results.json') for c in CONDITIONS
}
cross_results = _load(RESULTS / 'cross_condition_results.json')
shap_results = _load(RESULTS / 'shap_results.json')
noise_results = _load(RESULTS / 'noise_robustness.json', required=False)

gait_df = pl.read_csv(PROCESSED / 'gait_features.csv')

print(f'Within-condition loaded: {list(within_results)}')
print(f'Cross-condition directions: {list(cross_results)}')
print(f'SHAP directions: {list(shap_results)}')
print(f'Noise robustness present: {noise_results is not None}')
print(f'Gait feature matrix: {gait_df.shape}')
```

```
Within-condition loaded: ['pd', 'hd', 'als']
Cross-condition directions: ['pd_to_hd', 'hd_to_pd', 'pd_to_als', 'als_to_pd',
'hd_to_als', 'als_to_hd']
SHAP directions: ['als_to_pd', 'als_to_hd', 'pd_to_als', 'pd_to_hd',
'hd_to_als', 'hd_to_pd']
Noise robustness present: True
Gait feature matrix: (14753, 17)
```

1.1 Figure 1 — Within-condition overview

Composite 1×2 panel: (left) F1-macro heatmap across classifiers × conditions from Step 2 LOSO-CV; (right) modal-parameter frequency heatmap indicating how consistently the selected modal hyperparameters win across LOSO folds. A single-column variant contains only the F1 heatmap for a tight paper layout.

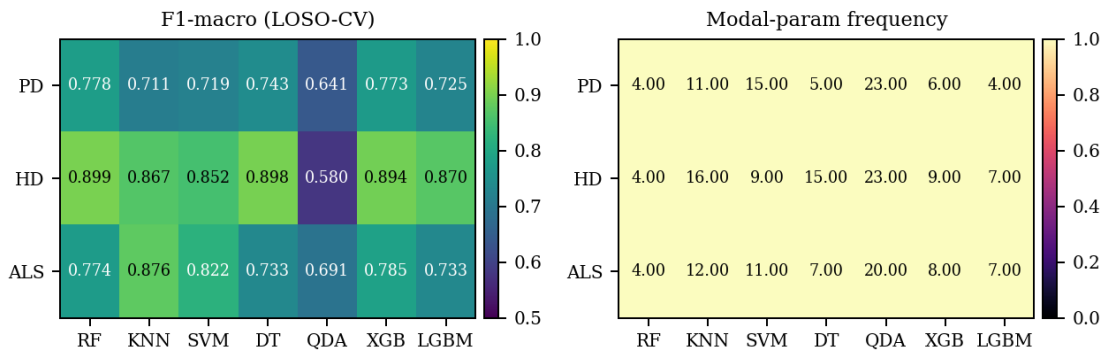
[3]:
```
f1_matrix = np.array([
    [within_results[c]['classifiers'][clf]['f1_macro'] for clf in CLASSIFIERS]
    for c in CONDITIONS
])
mf_matrix = np.array([
    [within_results[c]['classifiers'][clf].get('modal_frequency', np.nan) for␣
 ↪clf in CLASSIFIERS]
    for c in CONDITIONS
])

def _heatmap(ax, values, vmin, vmax, cmap, title, fmt='{:.3f}'):
    im = ax.imshow(values, vmin=vmin, vmax=vmax, cmap=cmap, aspect='auto')
    ax.set_xticks(range(len(CLASSIFIERS)))
    ax.set_xticklabels([CLF_LABEL[c] for c in CLASSIFIERS])
    ax.set_yticks(range(len(CONDITIONS)))
    ax.set_yticklabels([COND_LABEL[c] for c in CONDITIONS])
    ax.set_title(title)
    for i in range(values.shape[0]):
        for j in range(values.shape[1]):
            v = values[i, j]
            if np.isnan(v):
                continue
            bg = im.cmap(im.norm(v))[:3]
            lum = 0.299*bg[0] + 0.587*bg[1] + 0.114*bg[2]
            ax.text(j, i, fmt.format(v), ha='center', va='center',
                    fontsize=7, color='black' if lum > 0.55 else 'white')
    return im

fig, axes = plt.subplots(1, 2, figsize=(IEEE_DOUBLE, 2.4))
im1 = _heatmap(axes[0], f1_matrix, 0.5, 1.0, 'viridis', 'F1-macro (LOSO-CV)')
im2 = _heatmap(axes[1], mf_matrix, 0.0, 1.0, 'magma', 'Modal-param frequency',␣
 ↪fmt='{:.2f}')
fig.colorbar(im1, ax=axes[0], fraction=0.05, pad=0.02)
fig.colorbar(im2, ax=axes[1], fraction=0.05, pad=0.02)
fig.tight_layout()
save_paper(fig, 'fig1_within_overview')
plt.show()

fig, ax = plt.subplots(figsize=(IEEE_SINGLE, 2.0))
im = _heatmap(ax, f1_matrix, 0.5, 1.0, 'viridis', 'Within-condition F1-macro')
fig.colorbar(im, ax=ax, fraction=0.05, pad=0.02)
fig.tight_layout()
save_paper(fig, 'fig1_within_f1_single')
plt.show()
```

```
  wrote report/figures/paper/pdf/fig1_within_overview.pdf and
report/figures/paper/png/fig1_within_overview.png
```

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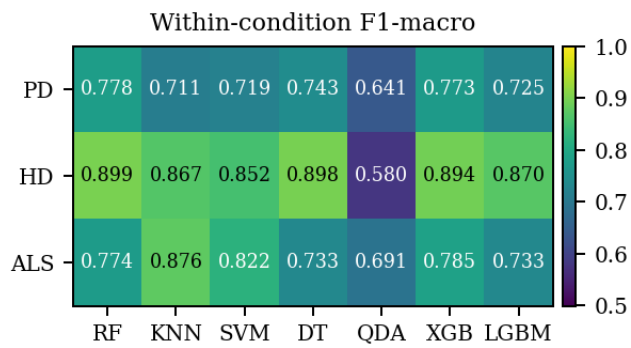
```



1.2 Figure 2 — Within-condition confusion matrices

Row of normalised confusion matrices for the modal-best classifier per condition (PD–RF, HD–RF, ALS–KNN, as established in Step 2). Single shared colourbar; no recomputation — plotted directly from the stored `y_true`/`y_pred` lists.

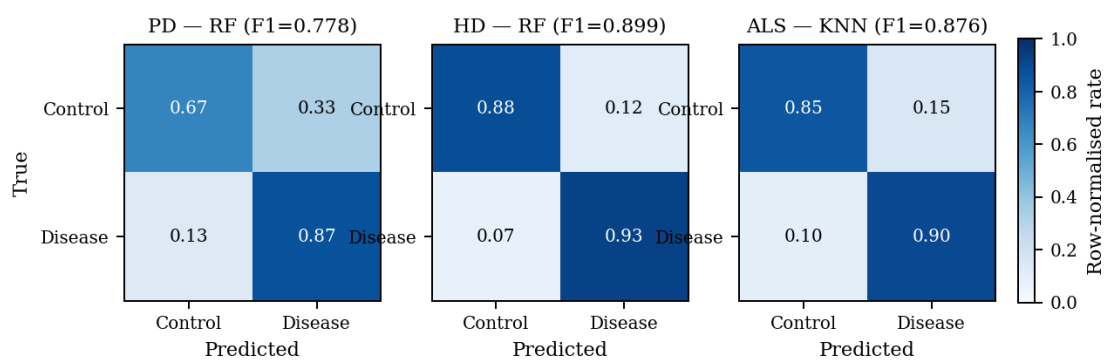
[4]:
```
BEST = [('pd', 'rf'), ('hd', 'rf'), ('als', 'knn')]

fig, axes = plt.subplots(1, 3, figsize=(IEEE_DOUBLE, 2.4))
ims = []
for ax, (cond, clf) in zip(axes, BEST):
    entry = within_results[cond]['classifiers'][clf]
    y_true = np.asarray(entry['y_true'])
    y_pred = np.asarray(entry['y_pred'])
    cm = confusion_matrix(y_true, y_pred, normalize='true')
    im = ax.imshow(cm, vmin=0, vmax=1, cmap='Blues')
    ims.append(im)
    ax.set_xticks([0, 1]); ax.set_yticks([0, 1])
```

```
    ax.set_xticklabels(['Control', 'Disease'])
    ax.set_yticklabels(['Control', 'Disease'])
    ax.set_xlabel('Predicted')
    if ax is axes[0]:
        ax.set_ylabel('True')
    ax.set_title(f'{COND_LABEL[cond]} - {CLF_LABEL[clf]} (F1={entry["f1_macro"]:
↪.3f})')
    for i in range(2):
        for j in range(2):
            v = cm[i, j]
            lum = 0.299 + 0.587*(1-v) + 0.114*(1-v)
            ax.text(j, i, f'{v:.2f}', ha='center', va='center',
                    fontsize=8, color='black' if v < 0.5 else 'white')

fig.subplots_adjust(right=0.9)
cbar_ax = fig.add_axes([0.92, 0.15, 0.015, 0.7])
fig.colorbar(ims[-1], cax=cbar_ax, label='Row-normalised rate')
save_paper(fig, 'fig2_within_cms')
plt.show()
```

```
wrote report/figures/paper/pdf/fig2_within_cms.pdf and
report/figures/paper/png/fig2_within_cms.png
```



1.3 Figure 3 — Cross-condition transfer

Top: per-classifier ΔF1 heatmap across the six transfer directions (positive = cross within, negative = degradation). Bottom: grouped within vs. cross F1 bars with stride-level 95% CIs. Both panels assembled from `cross_condition_results.json` (F1 and CI) and within-condition baselines in `{cond}_results.json` (F1 only).

[5]:
```
delta_f1 = np.zeros((len(CLASSIFIERS), len(DIRECTIONS)))
within_f1 = np.zeros_like(delta_f1)
cross_f1 = np.zeros_like(delta_f1)
cross_ci_lo = np.zeros_like(delta_f1)
```

```
cross_ci_hi = np.zeros_like(delta_f1)

for j, direction in enumerate(DIRECTIONS):
    src = direction.split('_to_')[0]
    for i, clf in enumerate(CLASSIFIERS):
        wf = within_results[src]['classifiers'][clf]['f1_macro']
        cf_entry = cross_results[direction]['classifiers'][clf]
        cf = cf_entry['f1_macro']
        within_f1[i, j] = wf
        cross_f1[i, j] = cf
        delta_f1[i, j] = cf - wf
        cross_ci_lo[i, j] = cf_entry.get('f1_macro_ci_lower', cf)
        cross_ci_hi[i, j] = cf_entry.get('f1_macro_ci_upper', cf)

fig, (ax_top, ax_bot) = plt.subplots(2, 1, figsize=(IEEE_DOUBLE, 5.2),
                                     gridspec_kw={'height_ratios': [1, 1.4]})

vmax = np.nanmax(np.abs(delta_f1))
im = ax_top.imshow(delta_f1, cmap='RdBu_r', vmin=-vmax, vmax=vmax,␣
 ↪aspect='auto')
ax_top.set_xticks(range(len(DIRECTIONS)))
ax_top.set_xticklabels([DIR_LABEL[d] for d in DIRECTIONS], rotation=0)
ax_top.set_yticks(range(len(CLASSIFIERS)))
ax_top.set_yticklabels([CLF_LABEL[c] for c in CLASSIFIERS])
ax_top.set_title('Transfer F1 shift  (ΔF1 = cross - within)')
for i in range(delta_f1.shape[0]):
    for j in range(delta_f1.shape[1]):
        v = delta_f1[i, j]
        ax_top.text(j, i, f'{v:+.2f}', ha='center', va='center',
                    fontsize=7, color='black' if abs(v) < vmax*0.5 else 'white')
fig.colorbar(im, ax=ax_top, fraction=0.04, pad=0.02)

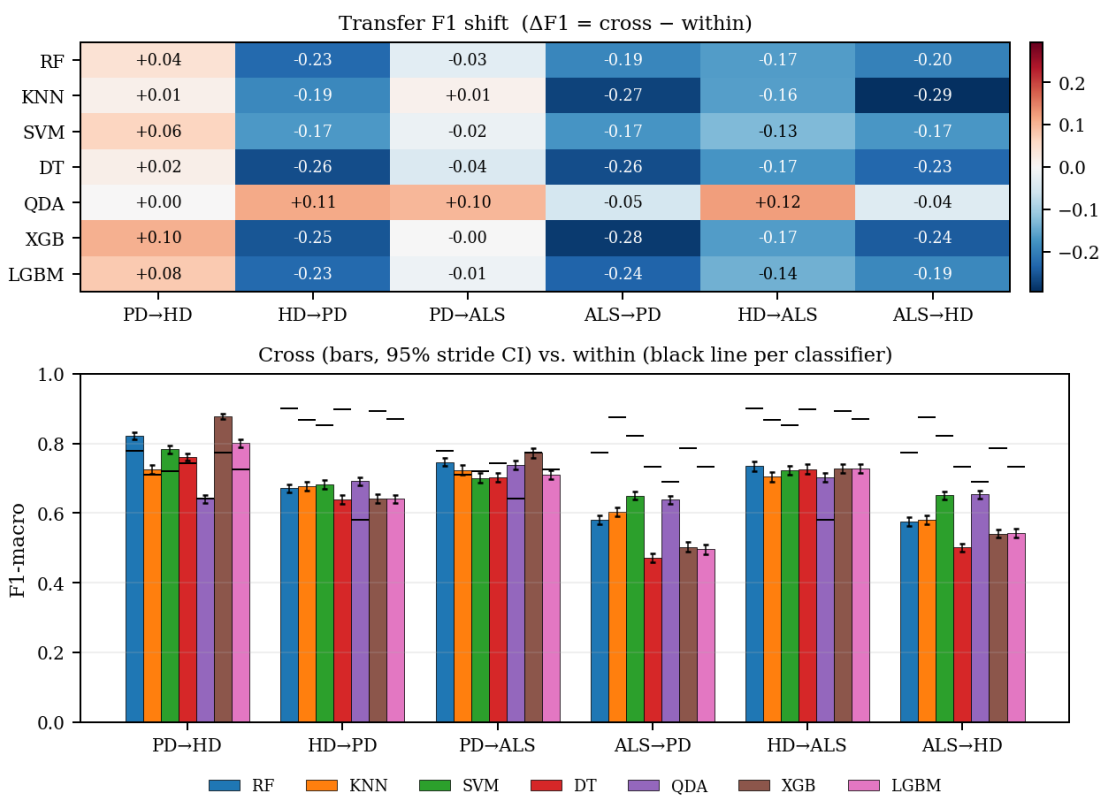
width = 0.8 / len(CLASSIFIERS)
x = np.arange(len(DIRECTIONS))
cmap_bar = plt.get_cmap('tab10')
for i, clf in enumerate(CLASSIFIERS):
    offs = (i - (len(CLASSIFIERS) - 1)/2) * width
    err_lo = cross_f1[i] - cross_ci_lo[i]
    err_hi = cross_ci_hi[i] - cross_f1[i]
    ax_bot.bar(x + offs, cross_f1[i], width=width,
               color=cmap_bar(i), edgecolor='k', linewidth=0.3,
               label=CLF_LABEL[clf],
               yerr=np.stack([err_lo, err_hi]), capsize=1.2,
               error_kw={'elinewidth': 0.6})
    for j in range(len(DIRECTIONS)):
        ax_bot.hlines(within_f1[i, j], x[j] + offs - width/2, x[j] + offs +␣
 ↪width/2,
```

```
                        colors='k', linewidths=0.8)

ax_bot.set_xticks(x)
ax_bot.set_xticklabels([DIR_LABEL[d] for d in DIRECTIONS])
ax_bot.set_ylabel('F1-macro')
ax_bot.set_ylim(0, 1.0)
ax_bot.set_title('Cross (bars, 95% stride CI) vs. within (black line per␣
 ↪classifier)')
ax_bot.legend(ncol=7, loc='upper center', bbox_to_anchor=(0.5, -0.12),␣
 ↪frameon=False)
ax_bot.grid(axis='y', alpha=0.2)

fig.tight_layout()
save_paper(fig, 'fig3_cross_degradation')
plt.show()
```

```
  wrote report/figures/paper/pdf/fig3_cross_degradation.pdf and
report/figures/paper/png/fig3_cross_degradation.png
```



1.4 Figure 4 — SHAP j diagnosis (Step 4)

Left: normalised j heatmap for RF across the six transfer directions, highlighting which features shift in attribution between within- and cross-condition predictions. Right: top-3 j features per direction as a compact table panel, extracted from `shap_results.json`.

[6]:
```
features = ALL_FEATURE_COLS

def _delta_map(direction, clf='rf', key='delta_j_normalized'):
    raw = shap_results[direction][clf][key]
    if isinstance(raw, dict):
        return {f: float(raw.get(f, np.nan)) for f in features}
    return {f: float(v) for f, v in zip(features, raw)}

delta_mat = np.zeros((len(features), len(DIRECTIONS)))
for j, direction in enumerate(DIRECTIONS):
    dj = _delta_map(direction)
    for i, feat in enumerate(features):
        delta_mat[i, j] = dj.get(feat, np.nan)

top3_rows = []
for direction in DIRECTIONS:
    dj = _delta_map(direction)
    ranked = sorted(dj.items(), key=lambda kv: -kv[1])[:3]
    top3_rows.append([DIR_LABEL[direction]] + [f'{f} ({v:.2f})' for f, v in␣
 ↪ranked])
top3_df = pd.DataFrame(top3_rows, columns=['Direction', 'Top-1', 'Top-2',␣
 ↪'Top-3'])

fig, (ax_hm, ax_tb) = plt.subplots(1, 2, figsize=(IEEE_DOUBLE, 4.2),
                                   gridspec_kw={'width_ratios': [1.2, 1.0]})

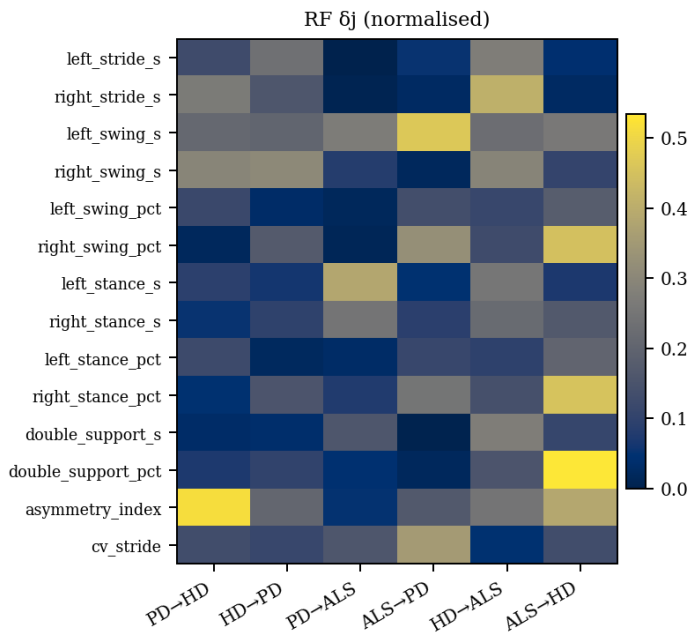
vmax = np.nanmax(delta_mat)
im = ax_hm.imshow(delta_mat, cmap='cividis', vmin=0, vmax=vmax, aspect='auto')
ax_hm.set_xticks(range(len(DIRECTIONS)))
ax_hm.set_xticklabels([DIR_LABEL[d] for d in DIRECTIONS], rotation=30,␣
 ↪ha='right')
ax_hm.set_yticks(range(len(features)))
ax_hm.set_yticklabels(features, fontsize=7)
ax_hm.set_title('RF  j (normalised)')
fig.colorbar(im, ax=ax_hm, fraction=0.04, pad=0.02)

ax_tb.axis('off')
table = ax_tb.table(cellText=top3_df.values, colLabels=list(top3_df.columns),
                    cellLoc='left', loc='center')
table.auto_set_font_size(False)
table.set_fontsize(7)
table.scale(1.0, 1.35)
```

```
ax_tb.set_title('Top-3  j features per direction')

fig.tight_layout()
save_paper(fig, 'fig4_shap_delta_j')
plt.show()
```

```
  wrote report/figures/paper/pdf/fig4_shap_delta_j.pdf and
report/figures/paper/png/fig4_shap_delta_j.png
```



Top-3 δj features per direction

Direction	Top-1	Top-2	Top-3
PD→HD	asymmetry	right_swing	right_stride_s
HD→PD	right_swing	left_stride_	asymmetry_in
PD→ALS	left_stance	left_swing_	right_stance_
ALS→PD	left_swing_	cv_stride ((	right_swing_
HD→ALS	right_stride	right_swing	double_suppo
ALS→HD	double_sup	right_stanc	right_swing_

1.5 Figure 5 — PCA + K-Means unified view (Step 6)

The PCA fit is reproduced here with Step-6-identical settings so the styling matches the rest of the paper. Left: PC1 vs. PC2 coloured by condition. Right: K-Means (K=3) cluster overlay. Bottom strip: cumulative explained variance.

[7]:
```
X = gait_df.select(ALL_FEATURE_COLS).to_numpy()
conditions_arr = gait_df['condition'].to_numpy()
labels_arr = gait_df['label'].to_numpy()

X_scaled = StandardScaler().fit_transform(X)
pca = PCA(n_components=len(ALL_FEATURE_COLS), random_state=42)
X_pca = pca.fit_transform(X_scaled)

kmeans = KMeans(n_clusters=3, random_state=42, n_init=20)
clusters = kmeans.fit_predict(X_scaled)
```

```
COND_ORDER = ['pd', 'hd', 'als', 'control']
COND_LABELS_FULL = {'pd': 'PD', 'hd': 'HD', 'als': 'ALS', 'control': 'Control'}
COND_COLORS_FULL = {'pd': '#1f77b4', 'hd': '#2ca02c', 'als': '#ff7f0e',␣
↪'control': '#7f7f7f'}

fig = plt.figure(figsize=(IEEE_DOUBLE, 4.0))
gs = fig.add_gridspec(2, 2, height_ratios=[4, 1], hspace=0.45, wspace=0.25)
ax_cond = fig.add_subplot(gs[0, 0])
ax_clust = fig.add_subplot(gs[0, 1])
ax_var = fig.add_subplot(gs[1, :])

for cond in COND_ORDER:
    mask = conditions_arr == cond
    if not mask.any():
        continue
    disease = (labels_arr == 1) & mask
    control = (labels_arr == 0) & mask
    ax_cond.scatter(X_pca[disease, 0], X_pca[disease, 1], s=6, alpha=0.35,
                    color=COND_COLORS_FULL[cond], marker='o',
                    label=f'{COND_LABELS_FULL[cond]}' if cond != 'control' else␣
↪None)
    ax_cond.scatter(X_pca[control, 0], X_pca[control, 1], s=10, alpha=0.45,
                    facecolors='none', edgecolors=COND_COLORS_FULL[cond],
                    linewidths=0.5, marker='o',
                    label='Control' if cond == 'control' else None)
ax_cond.set_xlabel('PC1'); ax_cond.set_ylabel('PC2')
ax_cond.set_title('PCA by condition')
ax_cond.legend(loc='best', frameon=False, fontsize=6, ncol=2)
ax_cond.grid(alpha=0.15)

scat = ax_clust.scatter(X_pca[:, 0], X_pca[:, 1], c=clusters, cmap='tab10',
                        s=6, alpha=0.45)
ax_clust.set_xlabel('PC1'); ax_clust.set_ylabel('PC2')
ax_clust.set_title('K-Means (K=3)')
ax_clust.grid(alpha=0.15)
legend = ax_clust.legend(*scat.legend_elements(), title='Cluster',
                         loc='best', fontsize=6, title_fontsize=6,␣
↪frameon=False)
ax_clust.add_artist(legend)

cum = np.cumsum(pca.explained_variance_ratio_)
comps = np.arange(1, len(cum) + 1)
ax_var.bar(comps, pca.explained_variance_ratio_, color='#4c72b0', alpha=0.7,
           label='Per-PC')
ax_var.plot(comps, cum, marker='o', ms=3, color='#dd8452', label='Cumulative')
ax_var.axhline(0.9, color='gray', linestyle='--', linewidth=0.6)
ax_var.set_xlabel('Principal component')
```

```
ax_var.set_ylabel('Variance ratio')
ax_var.set_xticks(comps)
ax_var.set_ylim(0, 1.02)
ax_var.legend(loc='upper left', frameon=False, fontsize=7)

save_paper(fig, 'fig5_pca_kmeans')
plt.show()
```

```
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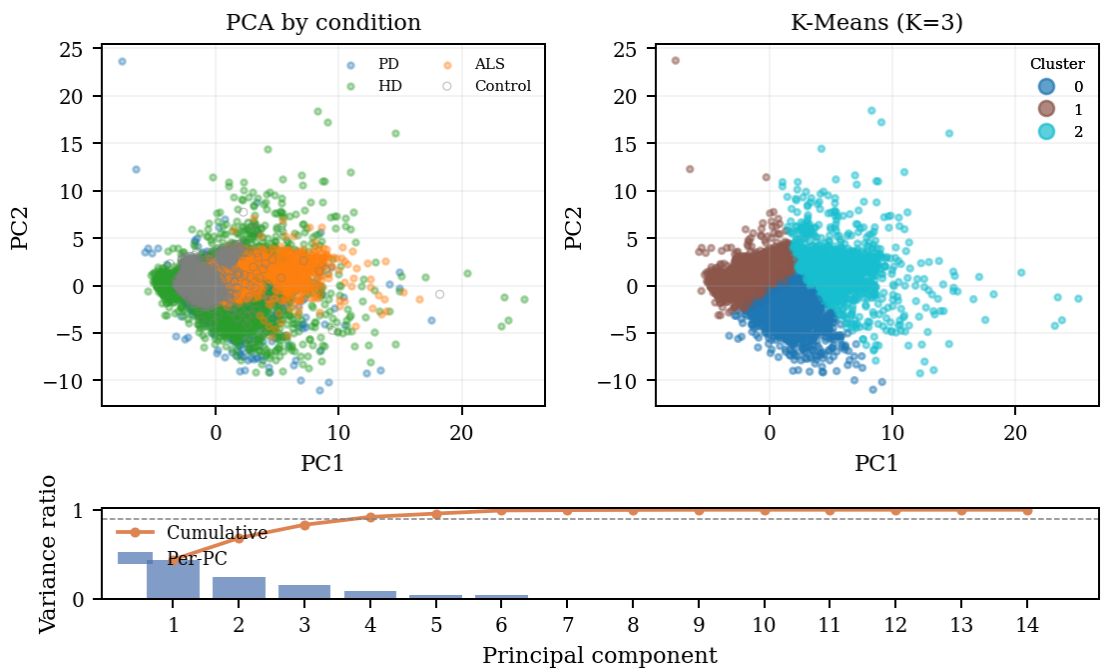
report/figures/paper/png/fig5_pca_kmeans.png
```



1.6 Figure 6 — Noise robustness (Step 5, within-condition)

F1-macro vs Gaussian input noise for each classifier, per condition. Clean baseline (=0) is a single point; {0.05…0.50} have 30 repeats — plotted as mean with ±1 SD shading. Cross-condition sweep is omitted (empty in the current `noise_robustness.json`).

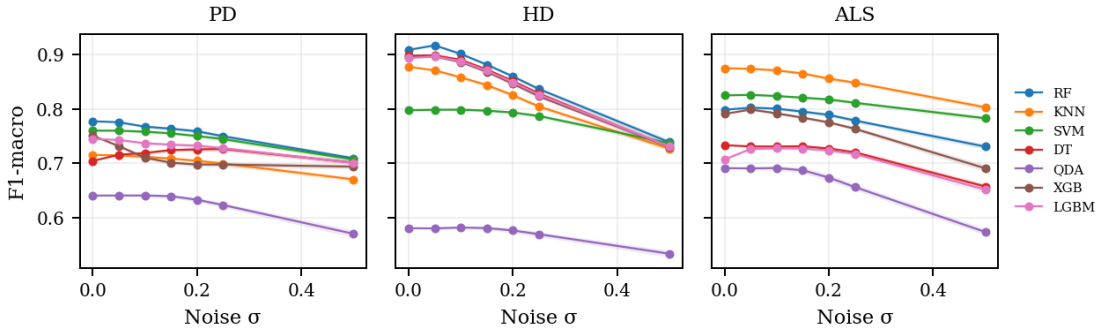
[8]:
```
if noise_results is None:
    print('noise_robustness.json missing - skipping Fig 6.')
else:
    sigma_keys = ['0.0', '0.05', '0.1', '0.15', '0.2', '0.25', '0.5']
    sigmas = np.array([float(s) for s in sigma_keys])

    fig, axes = plt.subplots(1, len(CONDITIONS), figsize=(IEEE_DOUBLE, 2.3),␣
 ↪sharey=True)
```

```
    cmap_line = plt.get_cmap('tab10')
    for ax, cond in zip(axes, CONDITIONS):
        for i, clf in enumerate(CLASSIFIERS):
            if clf not in noise_results['within'][cond]:
                continue
            per_sigma = noise_results['within'][cond][clf]
            means, stds = [], []
            for sk in sigma_keys:
                vals = np.asarray(per_sigma.get(sk, []), dtype=float)
                if vals.size == 0:
                    means.append(np.nan); stds.append(0.0)
                else:
                    means.append(vals.mean()); stds.append(vals.std(ddof=1) if␣
↪vals.size > 1 else 0.0)
            means = np.asarray(means); stds = np.asarray(stds)
            ax.plot(sigmas, means, marker='o', ms=3, linewidth=0.9,
                    color=cmap_line(i), label=CLF_LABEL[clf])
            ax.fill_between(sigmas, means - stds, means + stds,
                            color=cmap_line(i), alpha=0.15, linewidth=0)
        ax.set_xlabel('Noise σ')
        ax.set_title(COND_LABEL[cond])
        ax.grid(alpha=0.2)
    axes[0].set_ylabel('F1-macro')
    axes[-1].legend(loc='center left', bbox_to_anchor=(1.02, 0.5),
                    frameon=False, fontsize=6)
    fig.tight_layout()
    save_paper(fig, 'fig6_noise_robustness')
    plt.show()

    empty_cross = all(not noise_results['cross'][d] for d in␣
↪noise_results['cross'])
    if empty_cross:
        print('Cross-condition noise sweep is empty in noise_robustness.json;␣
↪not plotted.')
```

```
wrote report/figures/paper/pdf/fig6_noise_robustness.pdf and
report/figures/paper/png/fig6_noise_robustness.png
```



```
Cross-condition noise sweep is empty in noise_robustness.json; not plotted.
```

1.7 Master results table

Single long-form table with one row per (direction, classifier) carrying the within-condition F1 baseline, cross-condition F1 with both stride- and subject-level 95% CIs, permutation p-value, ΔF1, and the top-1 j feature (RF-normalised). Emitted to CSV and to an IEEE booktabs `tabular` for inclusion in the paper.

[9]:
```
def _delta_map_master(direction, clf='rf', key='delta_j_normalized'):
    raw = shap_results[direction][clf][key]
    if isinstance(raw, dict):
        return {f: float(raw.get(f, np.nan)) for f in ALL_FEATURE_COLS}
    return {f: float(v) for f, v in zip(ALL_FEATURE_COLS, raw)}

rows = []
for direction in DIRECTIONS:
    src = direction.split('_to_')[0]
    top1_feat = None
    if direction in shap_results:
        dj = _delta_map_master(direction)
        top1_feat = max(dj.items(), key=lambda kv: kv[1])[0]
    for clf in CLASSIFIERS:
        within_f1 = within_results[src]['classifiers'][clf]['f1_macro']
        cf_entry = cross_results[direction]['classifiers'][clf]
        rows.append({
            'direction': DIR_LABEL[direction],
            'classifier': CLF_LABEL[clf],
            'within_f1': round(within_f1, 4),
            'cross_f1': round(cf_entry['f1_macro'], 4),
            'delta_f1': round(cf_entry['f1_macro'] - within_f1, 4),
            'stride_ci_low': round(cf_entry.get('f1_macro_ci_lower', np.nan),␣
 ↪4),
            'stride_ci_high': round(cf_entry.get('f1_macro_ci_upper', np.nan),␣
 ↪4),
            'subj_ci_low': round(cf_entry.get('f1_macro_subj_ci_lower', np.
 ↪nan), 4),
            'subj_ci_high': round(cf_entry.get('f1_macro_subj_ci_upper', np.
 ↪nan), 4),
            'perm_p': round(cf_entry.get('permutation_p_value', np.nan), 4),
            'top1_delta_j_feature': top1_feat,
        })
master_df = pd.DataFrame(rows)
csv_path = TABLES / 'master_results.csv'
master_df.to_csv(csv_path, index=False)
```

```python
print(f'Wrote {csv_path.relative_to(Path("..."))}  ({len(master_df)} rows).')

def _tex_escape(s):
    if s is None:
        return '--'
    return str(s).replace('_', r'\_').replace('→', r'$\rightarrow$')

tex_lines = [
    r'% Auto-generated by notebooks/07_final_figures.ipynb',
    r'\begin{tabular}{llrrrrrrr}',
    r'\toprule',
    r'Direction & Clf & Within F1 & Cross F1 & $\Delta$F1 & Stride CI & Subj CI␣
↪& $p$ & Top-1 $\delta_j$ \\',
    r'\midrule',
]
current_dir = None
for _, r in master_df.iterrows():
    if r['direction'] != current_dir and current_dir is not None:
        tex_lines.append(r'\addlinespace')
    current_dir = r['direction']
    tex_lines.append(
        f"{_tex_escape(r['direction'])} & {_tex_escape(r['classifier'])} & "
        f"{r['within_f1']:.3f} & {r['cross_f1']:.3f} & {r['delta_f1']:+.3f} & "
        f"[{r['stride_ci_low']:.3f},{r['stride_ci_high']:.3f}] & "
        f"[{r['subj_ci_low']:.3f},{r['subj_ci_high']:.3f}] & "
        f"{r['perm_p']:.3f} & {_tex_escape(r['top1_delta_j_feature'])} \\\\"
    )
tex_lines += [r'\bottomrule', r'\end{tabular}']
tex_path = TABLES / 'master_results.tex'
tex_path.write_text('\n'.join(tex_lines) + '\n')
print(f'Wrote {tex_path.relative_to(Path("..."))}')

display(master_df.head(14))
```

```
Wrote report/tables/master_results.csv  (42 rows).
Wrote report/tables/master_results.tex
```

	direction	classifier	within_f1	cross_f1	delta_f1	stride_ci_low	\
0	PD→HD	RF	0.7777	0.8208	0.0431	0.8108	
1	PD→HD	KNN	0.7110	0.7260	0.0150	0.7135	
2	PD→HD	SVM	0.7194	0.7829	0.0636	0.7715	
3	PD→HD	DT	0.7434	0.7609	0.0174	0.7496	
4	PD→HD	QDA	0.6406	0.6414	0.0008	0.6296	
5	PD→HD	XGB	0.7727	0.8773	0.1046	0.8689	
6	PD→HD	LGBM	0.7251	0.8011	0.0760	0.7893	
7	HD→PD	RF	0.8995	0.6707	-0.2288	0.6582	
8	HD→PD	KNN	0.8669	0.6775	-0.1895	0.6656	
9	HD→PD	SVM	0.8524	0.6825	-0.1699	0.6696	

```
10     HD→PD          DT     0.8980    0.6382   -0.2599          0.6251
11     HD→PD         QDA     0.5799    0.6912    0.1114          0.6793
12     HD→PD         XGB     0.8937    0.6410   -0.2527          0.6283
13     HD→PD        LGBM     0.8703    0.6409   -0.2295          0.6285

    stride_ci_high  subj_ci_low  subj_ci_high  perm_p top1_delta_j_feature
0           0.8309       0.6676        0.9141     0.0      asymmetry_index
1           0.7384       0.5777        0.8310     0.0      asymmetry_index
2           0.7948       0.6161        0.8993     0.0      asymmetry_index
3           0.7721       0.5733        0.8851     0.0      asymmetry_index
4           0.6522       0.4923        0.7650     0.0      asymmetry_index
5           0.8858       0.7450        0.9600     0.0      asymmetry_index
6           0.8122       0.6195        0.9185     0.0      asymmetry_index
7           0.6831       0.4662        0.8451     0.0        right_swing_s
8           0.6899       0.4828        0.8270     0.0        right_swing_s
9           0.6950       0.4895        0.8436     0.0        right_swing_s
10          0.6516       0.4170        0.8240     0.0        right_swing_s
11          0.7030       0.5436        0.8098     0.0        right_swing_s
12          0.6534       0.4222        0.8222     0.0        right_swing_s
13          0.6528       0.4241        0.8234     0.0        right_swing_s
```

1.8 Verification

Sanity checks: every paper figure + both table files are present on disk, F1 values round-trip back to the source JSONs with machine precision, and the CSV row count matches the number of (direction × classifier) pairs.

[10]:
```
expected_figs = [
    'fig1_within_overview', 'fig1_within_f1_single', 'fig2_within_cms',
    'fig3_cross_degradation', 'fig4_shap_delta_j', 'fig5_pca_kmeans',
]
if noise_results is not None:
    expected_figs.append('fig6_noise_robustness')

for stem in expected_figs:
    for ext, d in [('pdf', PAPER_PDF), ('png', PAPER_PNG)]:
        assert (d / f'{stem}.{ext}').exists(), f'missing {d / (stem + "." +␣
 ↪ext)}'

for p in [TABLES / 'master_results.csv', TABLES / 'master_results.tex']:
    assert p.exists() and p.stat().st_size > 0, f'missing or empty: {p}'

reloaded = pd.read_csv(TABLES / 'master_results.csv')
assert len(reloaded) == len(DIRECTIONS) * len(CLASSIFIERS)

for direction in DIRECTIONS:
    src = direction.split('_to_')[0]
    for clf in CLASSIFIERS:
```

```
        row = reloaded[(reloaded['direction'] == DIR_LABEL[direction]) &
                       (reloaded['classifier'] == CLF_LABEL[clf])].iloc[0]
        json_within = within_results[src]['classifiers'][clf]['f1_macro']
        json_cross = cross_results[direction]['classifiers'][clf]['f1_macro']
        assert abs(row['within_f1'] - round(json_within, 4)) < 1e-6
        assert abs(row['cross_f1'] - round(json_cross, 4)) < 1e-6

print('All verifications passed.')
print(f'  {len(expected_figs)} figure stems × (pdf + png) present in report/
 ↪figures/paper/.')
print(f'  {len(reloaded)} master-table rows round-trip to JSON.')
```

```
All verifications passed.
  7 figure stems × (pdf + png) present in report/figures/paper/.
  42 master-table rows round-trip to JSON.
```